

Intermediate Task

Q. Perform sentiment analysis on X data to classify tweets as positive, negative, or neutral, and analyze trends.

Here is the dataset used for the X sentiment analysis:

<https://drive.google.com/file/d/15CMNw-F06CVAxcw4nAdc2LlsMKrmVVtT/view?usp=sharing>

```
# This code performs sentiment analysis on text data read from a CSV file named 'Tdata.csv'. It uses the VADER sentiment
```

```
# analyzer to compute sentiment scores for each sentence in the first column of the CSV. Based on the compound score,
```

```
# it classifies each sentence as Positive, Negative, or Neutral, and stores the results in a new DataFrame. Finally,
```

```
# The sentiment labels are saved to a new CSV file named 'sentiment_output.csv'.
```

```
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
import pandas as pd
```

```
df = pd.read_csv('sentiment_data.csv')
```

```
sid_obj = SentimentIntensityAnalyzer()
```

```
sentiment_results = []
```

```
for sentence in df.iloc[:, 0]:
```

```
    sentiment_dict = sid_obj.polarity_scores(sentence)
    compound = sentiment_dict['compound']
```

```
    if compound >= 0.05:
```

```
        sentiment = "Positive"
```

```
    elif compound <= -0.05:
```

```
        sentiment = "Negative"
```

```
    else:
```

```
        sentiment = "Neutral"
```

```
    sentiment_results.append(sentiment)
```

```
output_df = pd.DataFrame(sentiment_results, columns=['Sentiment'])
```

```
output_df.to_csv('sentiment_output.csv', index=False)
```

Here is the output file containing the sentiment classification results for the X sentiment analysis:

<https://drive.google.com/file/d/1-JSDrcOfPuUTWmLLmZQD9nyfSECAmkK7/view?usp=sharing>

X sentiment analysis Code for Graph:

```
import matplotlib.pyplot as plt
```

```
sentiment_counts= output_df['Sentiment'].value_counts().reindex(['Positive',  
'Negative'], fill_value=0)  
sentiment_counts.plot(kind='bar', color=['green', 'red'])  
plt.xlabel('Sentiment')  
plt.ylabel('Count')  
plt.title('Sentiment Analysis')  
plt.show()
```

Conclusion

The sentiment analysis using VADER shows a predominantly [Positive/Negative] sentiment trend based on the distribution of sentiment scores. Neutral comments were identified but not included in the visual representation. The analysis helps in understanding how people generally feel about the content in the dataset.

Output:-

